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Model-Free Sliding Mode Control for PMSM Based on Adaptive Super-Twisting Reaching Law

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Abstract-Permanent Magnet Synchronous Motors (PMSMs) are core actuators in electric drive systems for their high-power density and precision, but their control performance and robustness degrade severely under disturbances, abrupt load changes and parameter perturbations. Conventional Sliding Mode Control (SMC) has strong robustness but suffers from inherent high-frequency chattering and poor fixed-gain adaptability, failing to meet stringent high-precision electric demands. To address these issues, this paper proposes a novel model-free sliding mode control strategy (MF-TISMC-ADSTA-TISMO) integrating improved terminal integral sliding mode, adaptive super-twisting reaching law and improved Terminal Sliding Mode Observer (TISMO) for high-precision PMSM control. The core contributions are: 1). A novel PMSM ultra-local model incorporating state gains and lumped unknown disturbances is constructed, eliminating accurate motor model dependence and fundamentally eradicating model mismatch errors. 2). An improved terminal integral sliding mode surface with continuous smooth nonlinear functions is designed, which, combined with the adaptive super-twisting reaching law, ensures finite-time state convergence and mechanism-level chattering suppression. 3). An improved terminal sliding mode observer is developed to accurately estimate and compensate for total system disturbances in real time, significantly enhancing disturbance rejection under complex conditions. Experimental results demonstrate that the proposed strategy outperforms traditional methods in dynamic response, steady-state accuracy, multi-condition disturbance rejection and parameter robustness, validating its engineering value in high-performance electric drive control.

Keywords: Electric System; Permanent Magnet Synchronous Motor; Model-Free Sliding Mode Control; Adaptive Super-Twisting Reaching Law; Robustness

1 Introduction

PMSMs are widely employed as core actuators in modern electric drive systems due to their high power density, efficiency and excellent dynamic response [1-3], meeting the demands for power density, environmental adaptability and precise speed regulation. Their control performance directly determines the system's smooth starting, speed tracking, sudden load adaptation and stable under complex conditions. However, conventional SMC designs its sliding surface merely based on speed error and its derivative [Error! Reference

source not found.], resulting in inherent steady-state errors and high-frequency chattering under disturbances, load variations and parameter perturbations, which severely limits its application in high-precision electric systems [**Error! Reference source not found.**,6]. Its fundamental limitation is that the fixed-gain reaching law cannot adapt to complex disturbances, and it lacks a high-precision disturbance compensation mechanism.

Terminal integral sliding mode control (TISMC) has been developed to address steady-state errors. By introducing the terminal integral term into the sliding surface design, TISMC has been widely applied in PMSM control and effectively enhances steady-state precision and disturbance rejection [7,8]. However, in electric and other high-dynamic applications, large tracking errors easily cause integral saturation and deteriorated transient response in conventional TISMC, degrading system stability and restricting its practical use [9]. Existing solutions based on reducing integral gain or optimizing nonlinear parameters can relieve this issue but at the cost of response speed or robustness [10,11].

In terminal integral sliding mode control, the designs of the reaching law and disturbance observer directly determine the dynamic process of the system state approaching the sliding mode surface from initial conditions, and profoundly affect the steady-state chattering level and disturbance rejection capability after reaching the sliding mode surface [12]. The conventional first-order reaching law can ensure finite-time arrival at the sliding mode surface, but the discontinuous control signal it generates is the main source of high-frequency chattering. To suppress chattering fundamentally, second-order sliding mode control embeds the control action into higher-order derivatives, ensuring that the sliding mode variable and its first derivative can converge to zero in finite time, thereby generating continuous control signals. Among various second-order sliding mode algorithms, the super-twisting algorithm (STA) is highly favored due to its advantages of simple structure, no requirement for system acceleration information, and finite-time convergence [13,14]. However, its performance is highly dependent on the selection of algorithm gains: an excessively low gain fails to guarantee convergence, resulting in degraded robustness; an excessively high gain aggravates chattering and may even excite unmodeled dynamics [15]. Conventional gain tuning usually relies on conservative overestimation of the disturbance boundary, which poses a severe challenge in practical electric systems [16,17]. The randomness of disturbances, time-varying characteristics of loads and uncertainty of parameter perturbations make accurate or non-conservative disturbance boundary estimation difficult, so fixed gains often lead to insufficient robustness or excessive aggressiveness of the algorithm [18,19]. Meanwhile, conventional sliding mode observers suffer from low estimation accuracy and response lag for compound disturbances in electric drive systems, making it difficult to achieve accurate real-time compensation [20,21]. Model-free control has attracted extensive attention in the field of PMSM control due to its independence from the accurate mathematical model of the motor [22,23], but most existing model-free sliding mode schemes fail to organically combine the adaptive reaching law with high-precision disturbance observers, making them difficult to adapt to complex compound disturbance conditions [24]. This reveals that existing research still has obvious gaps in the organic integration of model-free frameworks, adaptive laws, and high-precision disturbance observers to

collaboratively resolve the trade-offs among chattering, dynamic response, and disturbance rejection in PMSM control.

In addition, model predictive control (MPC) serves as another important approach to enhance the robustness of PMSM drives. Ref. [28] optimizes finite control set model predictive current control with a sliding mode observer; Ref. [29] presents a simplified two-step MPC with fast voltage vector to reduce switching frequency; Ref. [30] applies model-free predictive control for harmonic suppression in PMSM systems. These methods belong to the MPC framework, which is technically different from the model-free sliding mode control scheme adopted in this paper.

Furthermore, within the model-free framework, Model-Free Predictive Control (MFPC) is another effective approach for high-performance SPMSM drives. Recent studies [31,32] have significantly improved current control and harmonic suppression in multiphase or SPMSM systems via optimized modulation, adaptive prediction horizons, or virtual harmonic back-EMF identification[33]. These works share this study's core tenet of eliminating accurate model dependence and focus on complex disturbance rejection. However, they differ fundamentally in technical paradigm: the aforementioned MFPC relies on predictive control's online optimization principle, while this paper proposes a synergistic scheme integrating an adaptive reaching law and improved observer based on sliding mode control's variable-structure robustness theory. This work provides a novel sliding-mode-based paradigm for high-performance, robust PMSM electric drive control. Therefore, to bridge this identified gap to meet the urgent requirements of PMSM electric drive systems for high-precision, strong disturbance-rejection and model-free control, a novel control strategy named MFITISMC-ADSTA-TISMO is proposed, which can collaboratively tackle multiple challenges including model mismatches, inherent chattering, disturbance estimation and dynamic response.

(1) An ultra-local model of PMSM is constructed by integrating state gains and lumped unknown disturbances, which fundamentally eliminates model mismatch errors and establishes the theoretical basis for model-free control.

(2) An improved terminal integral sliding-mode surface embedded with a continuous smooth function is developed. Combined with an adaptive super-twisting reaching law, it realizes chattering suppression at the mechanism level and finite-time convergence of system states.

(3) The improved terminal integral sliding mode observer (TISMO) enables accurate, real-time estimation and compensation of lumped unknown disturbances in electric drive systems. Its effectiveness is verified by experimental results on speed response, disturbance rejection and current smoothness, significantly strengthening the system's disturbance rejection and parametric robustness under complex conditions and sudden load changes.

2 Mathematical Model of the PMSM

Under ideal operating conditions, the mathematical equations of the PMSM are expressed as [25]:

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_f) \\ \frac{d\omega_e}{dt} = \frac{n_p}{J} (T_e - T_L - B\omega_m) \\ T_e = \frac{3}{2} n_p \psi_f i_q \end{cases} \quad (1)$$

where u_d and u_q denote the d-axis and q-axis stator voltages (V), respectively; i_d and i_q represent the d-axis and q-axis stator currents (A), respectively; R_s is the stator resistance (Ω); L_d and L_q are the d-q axis stator inductances (H); ω_e is the electric angular velocity (rad/s); n_p is the number of pole pairs; ω_m is the mechanical angular velocity (rad/s); B denotes the damping coefficient; J is the moment of inertia; T_L represents the load torque; T_e is the electromagnetic torque; where $\omega_e = n_p \omega_m$ defines the electric angular velocity; ψ_f is the permanent magnet flux linkage (Wb). Considering the impact of SPMSM operation under complex operating conditions on system stability, a further refined mathematical model of SPMSM is derived as:

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q + \Delta u_d \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_f) + \Delta u_q \\ \frac{d\omega_e}{dt} = \frac{n_p}{J} (T_e - T_L - B\omega_m) + \Delta P_n \\ T_e = \frac{3}{2} n_p \psi_f i_q + \Delta T_e \end{cases} \quad (2)$$

where Δu_d and Δu_q denote the d-axis and q-axis voltage disturbance components under complex operating conditions, respectively; ΔP_n denotes the disturbance arising from uncertainties or changes in the rotational inertia (J) and the damping coefficient (B). ΔT_e is the disturbance component of the output electromagnetic torque under complex operating conditions. Combining Eqs. (1) and (2) yields:

$$\frac{d\omega_e}{dt} = \left[\frac{3n_p}{2J} \psi_f i_q - \frac{B}{J} \omega_e + \frac{n_p}{J} (\Delta T_e - T_L + \Delta T_L) + \Delta P_n \right] \quad (3)$$

3 Design of the IFTSMC-ADSTA Controller and MFIFTSMC

3.1 Ultra-Local Model of the SPMSM Speed Loop

The ultra-local model of the single-input single-output (SISO) system can be expressed as [26]:

$$\begin{cases} y^{(n)} = h(x) + ku \\ y = x \end{cases} \quad (4)$$

where y is the system output; $y^{(n)}$ denotes the n th-order derivative of y with $n \geq 1$; $x \in \mathbb{R}$ is the system state; $h(x)$ is a bounded nonlinear Lipschitz function;

$k \in \mathbb{R}$ is a non-zero constant gain; and u is the control input. Based on the ultra-local model theory, the ultra-local model of the SPMSM speed loop can be established as:

$$\frac{d\omega_e}{dt} = \alpha i_q + f \quad (5)$$

where α is the system state gain; f represents the uncertainties of motor parameters and unknown disturbances, which can be estimated by an observer. The traditional model-free sliding mode control law for the speed loop can be derived from Eq. (10) as follows:

$$i_q^* = \frac{\dot{\omega}_e^* - f + u_c}{\alpha} \quad (6)$$

where ω_e^* is the derivative of the reference speed of the SPMSM; u_c is the sliding mode control input term. Substituting Eq. (5) into Eq. (6) yields:

$$u_c = -(\dot{\omega}_e^* - \dot{\omega}_e) \quad (7)$$

Taking the speed error as the state variable and differentiating it yields:

$$\begin{cases} \dot{x}_1 = e_1 = \omega_e^* - \omega_e \\ \dot{x}_1 = -u_c \end{cases} \quad (8)$$

where e_1 is the speed error. The sliding surface is chosen as:

$$s = C_1 e_1 + \dot{e}_1 \quad (9)$$

where $C_1 > 0$. The exponential reaching law is chosen as:

$$\dot{s} = -\varepsilon_1 \text{sign}(s) - k_1 s \quad (10)$$

where the parameters satisfy $\varepsilon_1 > 0$ and $k_1 > 0$. Figure 1 shows the block diagram of the SMC-ERL control. Substituting Eqs. (8) and (10) into Eq. (6) yields the traditional model-free sliding mode control law:

$$i_q^* = \frac{\dot{\omega}_e^* - f + C_1 e_1 + \dot{e}_1 + \varepsilon_1 \text{sign}(s) + k_1 s}{\alpha} \quad (11)$$

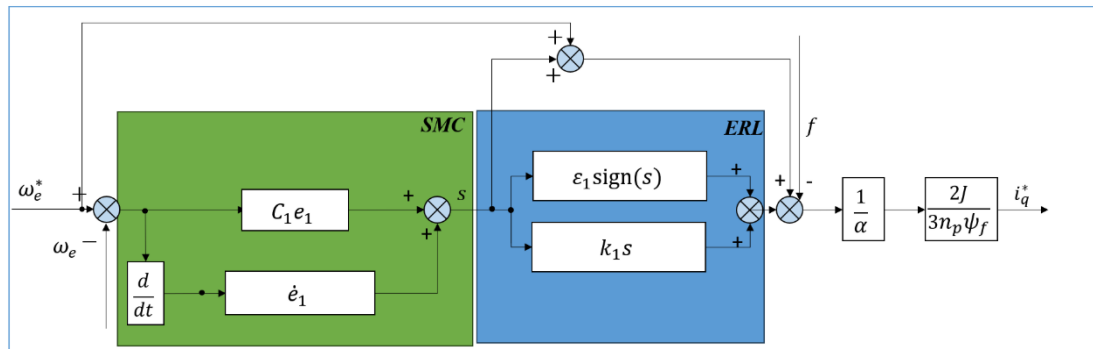


Figure 1. Block Diagram of the SMC-ERL Control

3.2 Design of MFITISM C

Considering parameter perturbations and external disturbances, and combining the advantages of MFC (model-free control) being independent of the motor model and ITISM C with reduced chattering and fast dynamic response, an MFITISM C speed-loop control method is proposed. This method enables the motor system to maintain high-precision control under dynamic operation. According to Ref. [26], $h(x)$ in Eq. (4) can be further decomposed into a linear state term and a nonlinear unknown disturbance term:

$$h(x) = \beta x + F \quad (12)$$

where β is the system state gain; F is the unknown disturbance term of the system, which contains the uncertain and certain parts of the motor parameters, and is a continuous nonlinear bounded function satisfying Lebesgue measurability. Equation (4) can be extended as:

$$\dot{x} = \alpha u + \beta x + F \quad (13)$$

Combining Eqs. (3) and (13), a novel ultra-local model of the speed loop can be established as:

$$\frac{d\omega_e}{dt} = \alpha i_q + \beta \omega_e + F \quad (14)$$

Based on Eq. (14), a sliding mode controller (SMC) based on the novel ultra-local model can be designed as:

$$i_q^* = \frac{\dot{\omega}_e^* - \beta \omega_e - F + u_c}{\alpha} \quad (15)$$

Defining $e_1 = \omega_e^* - \omega_e$ and substituting it into Eq. (15) yields:

$$\dot{\omega}_e^* - \dot{\omega}_e + u_c = 0 \quad (16)$$

Thus, we obtain:

$$u_c = -\dot{e}_1 \quad (17)$$

To effectively reduce the steady-state error and accelerate the system response speed, an improved terminal integral sliding surface is designed according to Ref. [27], as shown in Eq. (18).

$$\begin{cases} s = \dot{e}_1 + c_1 e_1 + c_2 \int_0^t y(e_1) dt \\ y(e_1) = \begin{cases} \gamma \cdot \frac{e_1}{\sqrt{e_1^2 + z}}, & |e_1| \geq \gamma \\ \gamma \cdot \tanh\left(\frac{e_1}{\gamma}\right), & |e_1| < \gamma \end{cases} \end{cases} \quad (18)$$

where $c_1 > 0$, $c_2 > 0$, $\gamma > 0$, and $0 < z < 1$. When the error is large ($|e_1| \geq \gamma$), the output of $\gamma \cdot \frac{e_1}{\sqrt{e_1^2 + z}}$ tends to be saturated. The growth of the integral term $c_2 \int_0^t y(e_1) dt$ is restricted by the boundedness of $y(e_1)$, leading to a slow accumulation rate. The proportional term $c_1 e_1$ increases linearly and dominates the sliding surface dynamics, driving the system to converge rapidly to the small-error region. When the error is small ($|e_1| < \gamma$), the integral term gradually becomes dominant by accumulating the smooth output of $y(e_1)$, which effectively eliminates the steady-state error. Meanwhile, the chattering of conventional sliding mode control is suppressed by the continuity of the $\tanh\left(\frac{e_1}{\gamma}\right)$ function, and the system robustness is guaranteed through the smooth transition of the piecewise function.

Theorem 1. For the speed error dynamics in Eq. (16), if the terminal integral sliding surface in Eq. (18) is chosen, the system states will converge rapidly in finite time after reaching the sliding surface.

Proof 1. After the system states reach the fast terminal sliding surface, it follows from the sliding mode equivalence principle that $s = \dot{s} = 0$. From Eq. (18), we obtain:

$$\dot{e}_1 = -c_1 e_1 - c_2 \int_0^t y(e_1) dt \quad (19)$$

The Lyapunov function is chosen as $V_1 = \frac{1}{2} e_1^2$.

$$\begin{aligned} \dot{V}_1 &= e_1 \dot{e}_1 \\ &= e_1 \left(-c_1 e_1 - c_2 \int_0^t y(e_1) dt \right) \\ &= -c_1 e_1^2 - c_2 \int_0^t e_1 y(e_1) dt \end{aligned} \quad (20)$$

From Eq. (20), it can be seen that $e_1 y(e_1) > 0$ when $|e_1| \geq \gamma$ and $e_1 y(e_1) > 0$ when $|e_1| < \gamma$. Thus, we can conclude that:

$$\dot{V}_1 \leq 0 \quad (21)$$

From Eq. (21), it can be concluded that the system states converge rapidly in finite time after reaching the sliding surface. Differentiating Eq. (18) and substituting the state equation (14) yields:

$$\begin{aligned} \dot{s} &= \ddot{e}_1 + c_1 \dot{e}_1 + c_2 y(e_1) \\ &= \ddot{e}_1 + c_1 (\dot{\omega}_e^* - \beta \omega_e - F + \alpha u_c) + c_2 y(e_1) \end{aligned} \quad (22)$$

Combining Eqs. (15) and (22) yields:

$$u_{eq} = \frac{1}{\alpha c_1} [\ddot{e}_1 + c_1 (\dot{\omega}_e^* - \beta \omega_e - F) + c_2 y(e_1)] \quad (23)$$

3.3 Design of the ADSTA

To drive the system states to reach the terminal sliding surface rapidly, an adaptive super-twisting reaching law is designed, as shown in Eq. (24).

$$\begin{cases} \dot{s} = -k_1(t)|s|^{\frac{1}{2}}\text{sign}(s) + g + \rho \\ \dot{g} = -k_2(t)\text{sign}(s) \\ \dot{k}_1(t) = \eta_1|s|^{\frac{1}{2}} - \sigma_1 k_1(t) \\ \dot{k}_2(t) = \eta_2[k_1(t)]^2 + \sigma_2 s \end{cases} \quad (24)$$

where $\eta_1 > 0$, $\sigma_1 > 0$, $\eta_2 > 0$, $\sigma_2 > 0$. The time-varying disturbance term ρ is contained in the lumped disturbance term F . When the system states are far from the sliding surface ($|s|$ is large), the term $k_1(t)|s|^{\frac{1}{2}}\text{sign}(s)$ and the rapidly increasing $k_1(t)$ dominate together, providing a strong driving effect to achieve fast convergence. When the states approach the sliding surface ($|s|$ is small), the disturbance estimation value g accumulated previously via $\dot{g} = -k_2(t)\text{sign}(s)$ plays a dominant compensation role, which effectively cancels the lumped disturbance, while the gains $k_1(t)$ and $k_2(t)$ tend to be stable. Ultimately, the inherent chattering problem of conventional sliding mode control is significantly suppressed while ensuring finite-time convergence and strong robustness.

Next, the finite-time convergence of the system is proved. Figure 2 shows the phase trajectories of each control algorithm under disturbance-free and linearly increasing disturbances, which intuitively compares their convergence dynamics and disturbance rejection performance.

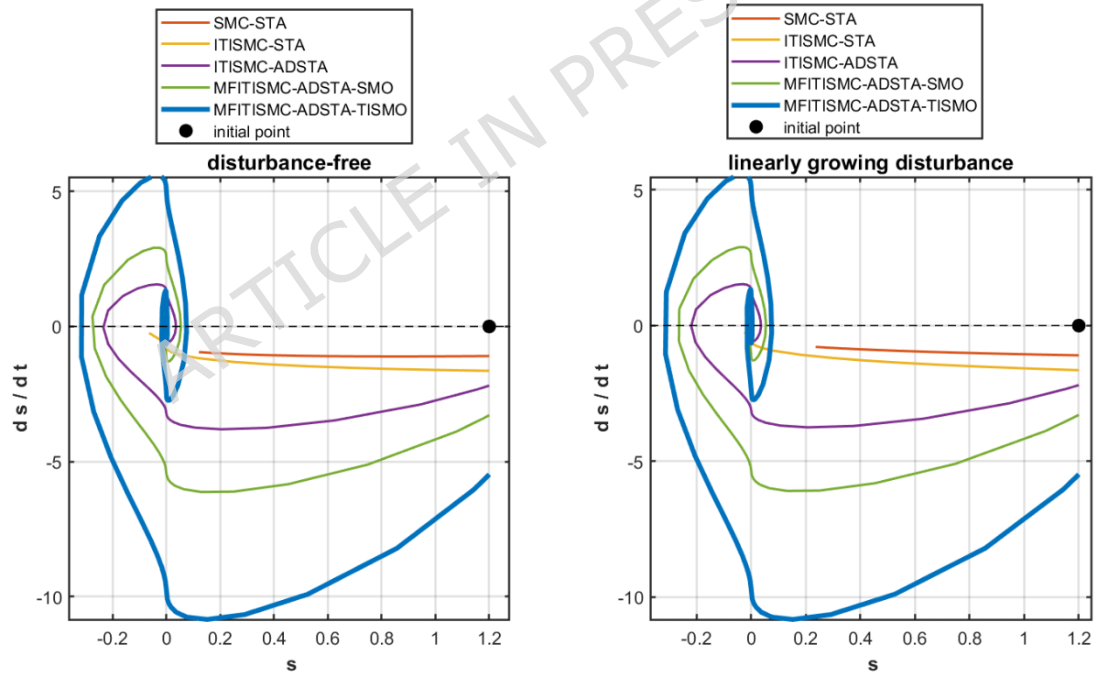


Figure 2. Phase trajectory diagram

Proof 2. The quadratic Lyapunov function is chosen as:

$$V_2 = \frac{1}{2}s^2 + \frac{1}{2}g^2 \quad (25)$$

where $V_1 > 0$ for $(s,g) \neq (0,0)$ and $V_1 = 0$ if and only if $s = g = 0$ (equilibrium state). From Eq. (24), we have $\dot{k}_1(t) = \eta_1|s|^{\frac{1}{2}} - \sigma_1 k_1(t)$, whose general solution is:

$$k_1(t) = e_1^{-\sigma_1 t} \left[k_1(0) + \eta_1 \int_0^t e_1^{\sigma_1 \tau} |s(\tau)|^{1/2} d\tau \right] \quad (26)$$

As $s \rightarrow 0$, the integral term $\int_0^t e_1^{\sigma_1 \tau} |s(\tau)|^{1/2} d\tau \rightarrow 0$ (since s decays to zero). Thus, $k_1(t)$ is dominated by the exponential decay term $e_1^{-\sigma_1 t}$. There exists a finite time $T_1 > 0$ such that $k_1(t) \leq \rho$ for $t \geq T_1$, where $\rho > 0$ is an arbitrarily small positive constant determined by the initial value $k_1(0)$ and the decay rate of s . Differentiating Eq. (25):

$$\dot{V}_2 = s\dot{s} + g\dot{g} \quad (27)$$

Substituting Eq. (24) into Eq. (27) and using $s \cdot \text{sign}(s) = |s|$ yields:

$$\dot{V}_2 = -k_1(t)|s|^{3/2} + sg + s\rho - \eta_2[k_1(t)]^2 g \cdot \text{sign}(s) - \sigma_2 s^2 \quad (28)$$

For $t \geq T_1$, $k_1(t) \leq \rho \ll 1$, and thus $k_2(t) = \eta_2[k_1(t)]^2 + \sigma_2 s$ (where $k_1(t)$ can be neglected). The integral term of $g(t)$ contains $k_2(t)\text{sign}(s)$, and as $k_2(t)$ decays, $g(t)$ also tends to zero (due to the convergence of the integral). From Eq. (28), we obtain:

$$\begin{aligned} s\dot{s} &= -k_1(t)|s|^{3/2} + s(g + \rho) \\ &\leq -k_{\min}|s|^{3/2} \end{aligned} \quad (29)$$

where $k_{\min} > 0$ is the lower bound of $k_1(t)$. Near the sliding surface, $V_2 = \frac{1}{2}s^2$ ($g \rightarrow 0$). Combining Eqs. (27)-(29) and using $|s| = \sqrt{2V_2}$, we obtain:

$$\dot{V}_2 \leq -2^{3/4}k_{\min}V_2^{3/4} \quad (30)$$

Let $\theta = 2^{3/4}k_{\min}$ and $\kappa = \frac{3}{4} \in (0,1)$. Then we obtain:

$$\dot{V}_2 \leq -\theta V_2^\kappa \quad (31)$$

Separating variables on both sides of Eq. (31) and integrating them, the convergence time T_2 is obtained as:

$$T_2 \leq \frac{V(0)^{1-\kappa}}{\theta(1-\kappa)} \quad (32)$$

According to the finite-time Lyapunov theorem, $V_2(t)$ will decrease from $V_2(0)$ to 0 in finite time, which further drives s and g to converge to 0 simultaneously. Q.E.D.

Combining Eqs. (23) and (24) yields the overall MFITISM control law, as shown in Eq. (33).

$$u_c = \left[\begin{aligned} &\frac{1}{\alpha c_1} [\ddot{e}_1 + c_1(\dot{\omega}_e^* - \beta \omega_e - F) + c_2 y(e_1)] \\ &+ k_1(t)|s|^{\frac{1}{2}} \text{sign}(s) + \int k_2(t) \text{sign}(s) dt \end{aligned} \right] \quad (33)$$

The block diagram of the proposed MFITISM algorithm is shown in Fig. 3.

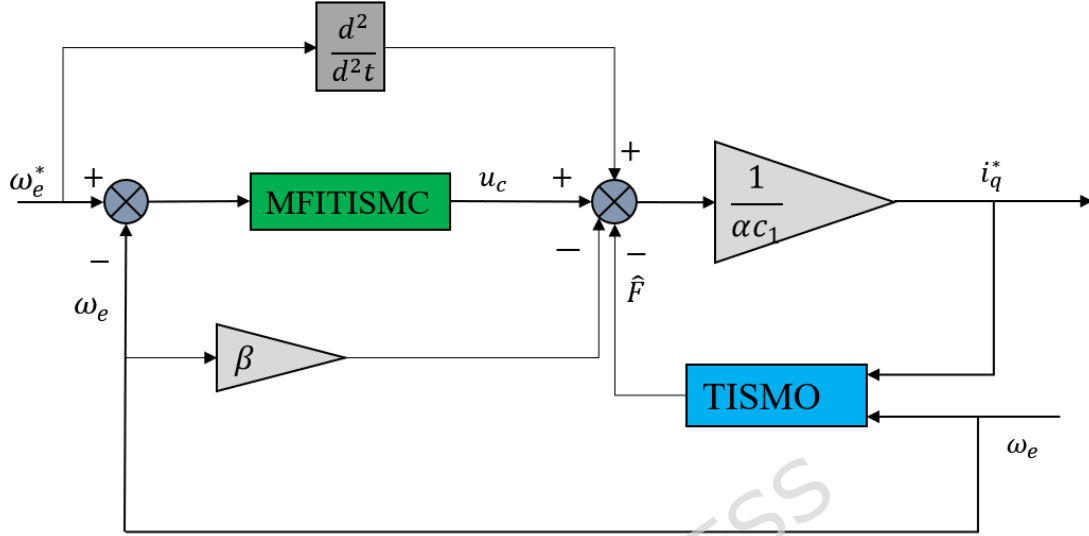


Fig. 3 Block diagram of the MFITISM

4 Design of the TISMO

From Eq. (14), there exists an unknown disturbance F in the system control input. By extending F as a state variable, an improved fast terminal sliding mode observer can be designed to accurately estimate the unknown lumped disturbance. From Eq. (14), the extended novel ultra-local model of the speed loop is obtained as:

$$\begin{cases} \frac{d\omega_e}{dt} = \alpha i_q + \beta \omega_e + F \\ \frac{dF}{dt} = \varphi(t) \end{cases} \quad (34)$$

where $\varphi(t)$ denotes the rate of change of the unknown disturbance. For the system described by Eq. (34), an improved fast terminal sliding mode observer is designed, as shown in Eq. (35).

$$\begin{cases} \frac{d\hat{\omega}_e}{dt} = \alpha i_q + \beta \hat{\omega}_e + \hat{F} + u_{smo} \\ \frac{d\hat{F}}{dt} = \xi \cdot u_{smo} \end{cases} \quad (35)$$

where $\hat{\omega}_e$ is the observed speed, $\hat{F}$ is the real-time observation of the unknown part of the system, u_{smo} is the sliding mode control law, and $\xi > 0$ is the observer gain. Combining Eqs. (34) and (35) yields the speed observation error equation, as shown in Eq. (36).

$$\begin{cases} \dot{e}_\omega = \beta e_\omega + e_f + u_{smo} \\ \dot{e}_f = \xi \cdot u_{smo} - \varphi(t) \end{cases} \quad (36)$$

where $e_\omega = \hat{\omega}_e - \omega_e$ is the speed estimation error, and $e_f = \hat{F} - F$ is the observation error of the unknown part. For the TISMO observer, the sliding surface is chosen as Eq. (37), which is given by Eq. (18).

$$\begin{cases} s = \dot{e}_\omega + c_1 e_\omega + c_2 \int_0^t y(e_\omega) dt \\ y(e_\omega) = \begin{cases} \gamma \cdot \frac{e_\omega}{\sqrt{e_\omega^2 + z}}, & |e_\omega| \geq \gamma \\ \gamma \cdot \tanh\left(\frac{e_\omega}{\gamma}\right), & |e_\omega| < \gamma \end{cases} \end{cases} \quad (37)$$

The sliding mode reaching law is chosen as:

$$\dot{s} = -N \text{sign}(s) \quad (38)$$

where $N > 0$.

Theorem 3. Choosing Eq. (37) as the sliding surface and Eq. (38) as the sliding-mode reaching law, the system is finite-time stable for the parameter $N > 0$.

Proof 3. Choose the Lyapunov function $V_3 = \frac{1}{2}s^2$ and take its time derivative:

$$\begin{aligned} \dot{V}_3 &= s\dot{s} \\ &= s[-N \text{sign}(s)] \\ &= -W|s| \\ &\leq 0 \end{aligned} \quad (39)$$

With the parameter $N > 0$, it can be guaranteed that the system states reach the sliding surface in finite time. From Eqs. (37)–(39), the control law of the TISMO is obtained as:

$$u_{smo} = c_1 \dot{e}_1 + c_2 y(e_1) - \beta \dot{e}_\omega + N \text{sign}(s) \quad (40)$$

From Eqs. (40) and (35), $\hat{F}$ is derived as:

$$\hat{F} = \varepsilon \int_0^t c_1 \dot{e}_1 + c_2 y(e_1) - \beta \dot{e}_\omega + N \text{sign}(s) dx \quad (41)$$

Substituting Eq. (41) into Eq. (35) yields the total control law i_q^* of the controller:

$$\begin{aligned}
i_q^* &= \frac{\hat{\omega}_e^* - \lambda_2 \hat{\omega}_e - \hat{F} + u_{smo}}{\alpha} \\
&= \frac{1}{\alpha} \left[\hat{\omega}_e^* + c_1 \dot{e}_1 + c_2 y(e_1) - \beta \dot{e}_\omega \right. \\
&\quad \left. + N \text{sign}(s) - \lambda_2 \hat{\omega}_e - \hat{F} \right]
\end{aligned} \tag{42}$$

The above design ensures that the TISMO can rapidly track and compensate for lumped disturbances within finite time.

5 Experimental Analysis

5.1 Experimental Setup

To verify the effectiveness and superiority of the proposed MFITISMC-ADSTA-TISMO controller in this paper, an experimental platform is established and physical experiments are implemented. The experimental platform is illustrated in Fig. 4, which employs the TMS320F28379D DSP chip. It consists of a DSP control platform, a PMSM, a dynamic torque sensor, and a two-level voltage-source three-phase inverter based on DRV8323H, forming the hardware architecture of a dual-inverter motor back-to-back test bench. Comparative experiments are carried out between the proposed controller and SMC-STA, ITISMC-STA, ITISMC-ADSTA, SMC[26], and MFITISMC-ADSTA-SMO. The motor parameters and the parameters of the five controllers are listed in Table 1 and Table 2, respectively. The vector control system adopts space vector pulse width modulation (SVPWM) with a switching frequency of 10 kHz. A low-pass filter is utilized with a cutoff frequency of 0.001.

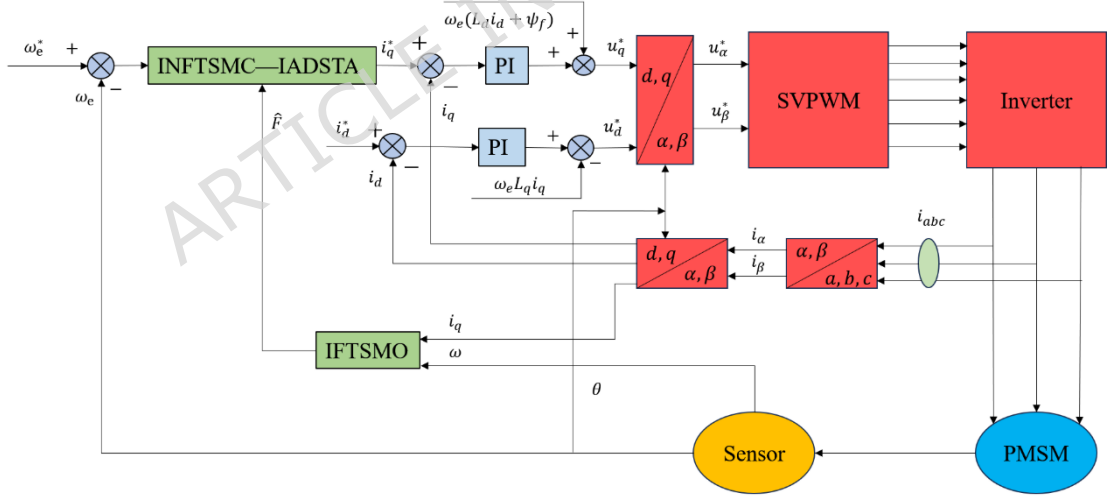


Figure 4. MFITISMC-ADSTA-TISMO Control Block Diagram

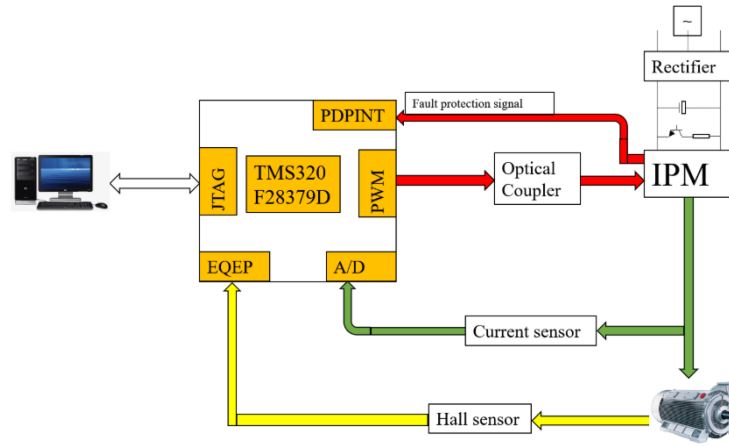


Figure 5. Experimental Setup

To evaluate the real-time performance of the proposed control strategy, an experimental platform based on TMS320F28379D DSP (200 MHz) and DRV8323H gate driver is established. With PWM and control interrupt frequencies set to 10 kHz (100 μ s period), all computations (current sampling, coordinate transformation, MFITISMC law, adaptive super-twisting reaching law, TISMO iteration, SVPWM) are completed. Despite higher complexity than conventional SMC controllers (additional adaptive gain and observer computations), the increased operations are within the DSP's real-time capacity. All experiments (load step change, sinusoidal disturbance, back-to-back tests) run stably at 10 kHz without interrupt overflow or instability, experimentally verifying the MFITISMC-ADSTA-TISMO algorithm's feasibility and real-time performance on hardware.

The experimental conditions are set as follows: the motor starts under no-load condition with a given speed of 600 rpm. The speed is raised to 900 rpm at 10 s, then reduced to 750 rpm at 20 s. A load torque of 0.1 N·m is applied at 25 s and removed at 30 s. During 35–40 s, the frequency of the sinusoidal disturbance is set to $0.18\tau_m^{-1}$, where τ_m represents the mechanical time constant. The flux linkage is adjusted to 1.3 times its initial value at 50 s. The corresponding speed waveform and q-axis current waveform are shown in Fig. 6 and Fig. 7, respectively.

Table 1. Parameters of SPMSM

parameters	numerical value
Number of pole pairs	4
Stator resistance	0.1 Ω
Stator inductance (d-q axis)	1.9×10^{-4} H
Permanent magnet flux linkage	0.0133 Wb
Moment of inertia	4.03×10^{-4} kg·m ²
Viscous damping coefficient	3.1136×10^{-4} N·m·s

To comprehensively evaluate the proposed strategy, comparative experiments are conducted with six control schemes: SMC-STA, ITISMC-STA, ITISMC-ADSTA, Model-Free Improved Terminal Integral Sliding MFITISMC-ADSTA-SMO, the method in Reference [30], and the proposed MFITISMC-ADSTA-TISMO. Detailed controller parameters are given in Table 2.

To ensure engineering practicality and reproducibility of the proposed control strategy, a stepwise experimental-based parameter tuning procedure is presented.

All parameters are finalized on the identical experimental platform (Section 5.1) by comparing system dynamic responses to standard test cases: speed step, sudden load torque change, and sinusoidal disturbance. Tuning follows the principle: 1). Tune core parameters of the . Fixing other module gains, adjust ITISMS parameters to establish baseline dynamic response and steady-state accuracy. 2). Tune parameters of the . With ITISMS parameters fixed, adjust adaptive gains to optimize the tradeoff between convergence speed and chattering suppression. 3). Tune parameters of the . After main controller performance meets specifications, adjust ITSMO gains to enhance lumped disturbance estimation, maximizing feedforward compensation effectiveness.

Table 2. Parameters of the Controllers

SMC—STA	$c_1 = 8; \varepsilon_1 = 2600; k_1 = 12000$
ITISMC—STA	$c_1 = 8; \varepsilon_1 = 2200; k_1 = 15000;$ $c_2 = 5; \gamma = 10^{-6}; z = 14000$
ITISMC—ADSTA	$c_1 = 8; \eta_1 = 0.5; \sigma_1 = 0.005; \eta_2 = 0.02; \sigma_2$ $= 0.006;$ $c_2 = 0.3; \gamma = 0.8; z = 0.05$
MFITISMC—ADSTA—SMO	$c_1 = 5.5; N = 11000$
SMC[26]	$\eta_1 = 8; \eta_2 = 0.003; p = 5, q = 7; \varepsilon = 17000 ;$ $\lambda = 12$
MFITISMC—ADSTA—TISMO	$c_1 = 5.5; \eta_1 = 0.5; \sigma_1 = 0.006; \eta_2 = 0.03; \sigma_2$ $= 0.004;$ $c_2 = 0.25; \gamma = 0.7; z = 0.06; N = 14000$

(Note: Control law in Ref. [30]: $\dot{i}_q^* = \frac{-F + \dot{\omega}_e^*}{\alpha} + \frac{\varepsilon \operatorname{sgn}(s) + \lambda s + \ddot{e} + \eta_2 |e|^{p/q} \operatorname{sgn}(e)}{\alpha \cdot \eta_1}$)

5.2 Experimental Results and Performance Analysis

1. Starting Condition:

As shown in Fig. 6(a), during the motor's no-load start-up to 600 r/min, the SMC—STA algorithm takes 0.85 s. The fixed-gain reaching law in conventional SMC causes slow transient convergence, sluggish speed rise and obvious hysteresis. ITISMC—STA shortens the start-up time to 0.75 s via the finite-time convergence of the terminal integral sliding surface, improving system transient response. ITISMC—ADSTA further reduces it to 0.7 s by adaptively adjusting gains with ADSTA according to the start-up state, significantly easing speed hysteresis. MFITISMC—ADSTA—SMO optimizes the time to 0.6 s through model-free design and SMO disturbance compensation, accelerating speed tracking while lowering model dependence. The proposed MFITISMC—ADSTA—TISMO completes start-up in only 0.55 s (35.2% faster than SMC—STA). Its ultra-local model integrates state gains and disturbance characteristics; combined with precise TISMO observation and ADSTA adaptive adjustment, it achieves steep, overshoot- and hysteresis-free speed rise and rapid entry into steady state.

2. Dynamic Speed-Up:

As shown in Fig. 6(b), the system steps speed from 600 r/min to 900 r/min at 10 s. SMC—STA takes 0.9 s for speed-up regulation, with low tracking accuracy, obvious fluctuations and climbing stagnation due to its fixed gain failing to adapt to operating condition changes. ITISMC—STA enhances speed tracking and shortens the time to 0.75 s, reducing speed fluctuations via the terminal integral

sliding surface. ITISM-C-ADSTA compresses the regulation time to 0.65 s with real-time ADSTA time-varying gain matching, accelerating sliding surface approach and eliminating speed stagnation. MFITISM-C-ADSTA-SMO optimizes it to 0.55 s via model-free design and SMO disturbance compensation, realizing smooth speed transition. MFITISM-C-ADSTA-TISMO finishes regulation in just 0.5 s (44.4% faster than SMC-STA), achieving overshoot- and stagnation-free real-time speed tracking thanks to the improved ultra-local model, TISMO precise compensation and ADSTA adaptation.

3. Steady-State Operation

As shown in Fig. 6(c), at a steady 900 r/min, SMC-STA has a speed fluctuation of 7.5 r/min, with high-frequency chattering and obvious periodic ripples caused by the sign function and lack of disturbance compensation. ITISM-C-STA reduces the fluctuation to 6.5 r/min by improving the sign function. ITISM-C-ADSTA lowers it to 5 r/min via ADSTA adaptive gain damping small oscillations. MFITISM-C-ADSTA-SMO further cuts it to 4 r/min through SMO steady-state disturbance observation and compensation. MFITISM-C-ADSTA-TISMO achieves the lowest fluctuation of 3 r/min (60% lower than SMC-STA). It suppresses chattering with a smooth nonlinear function, and TISMO plus the ultra-local model provide accurate disturbance compensation, delivering an almost flat speed curve and optimal steady-state control accuracy.

4. Dynamic Process of Speed Reduction

As shown in Fig. 6(d), the system steps speed from 900 r/min to 750 r/min at 20 s. SMC-STA has a 9 r/min speed undershoot and 0.85 s settling time, prone to impulse responses due to no integral characteristics and fixed gain mismatch. ITISM-C-STA reduces undershoot to 8 r/min and settling time to 0.75 s via the integral property of the terminal integral sliding surface. ITISM-C-ADSTA further cuts undershoot to 5 r/min and time to 0.7 s with ADSTA adaptive gain matching. MFITISM-C-ADSTA-SMO achieves negligible undershoot with a 0.6 s settling time through model-free design and SMO compensation. MFITISM-C-ADSTA-TISMO realizes zero overshoot/undershoot in only 0.55 s, ensuring smooth speed-down and fast steady recovery via enhanced fluctuation suppression and TISMO-ADSTA real-time compensation.

5. Anti-Disturbance Performance Under Step Change

As shown in Figs. 6(e) and 6(f), a 0.1 N·m load torque is applied at 25 s and removed at 30 s. SMC-STA has a 150 r/min speed fluctuation with sharp changes from lacking disturbance compensation. ITISM-C-STA reduces it to 128 r/min via finite-time convergence. ITISM-C-ADSTA optimizes it to 113 r/min by adaptively offsetting disturbance-induced fluctuations. MFITISM-C-ADSTA-SMO lowers it to 105 r/min through SMO load disturbance compensation. MFITISM-C-ADSTA-TISMO has only 85 r/min fluctuation (43.3% lower than SMC-STA), producing tiny speed variations and rapid steady recovery via TISMO real-time compensation, ADSTA residual disturbance offsetting and model-free design.

6. Sinusoidal Disturbance Condition with $0.18 \tau_m^{-1}$

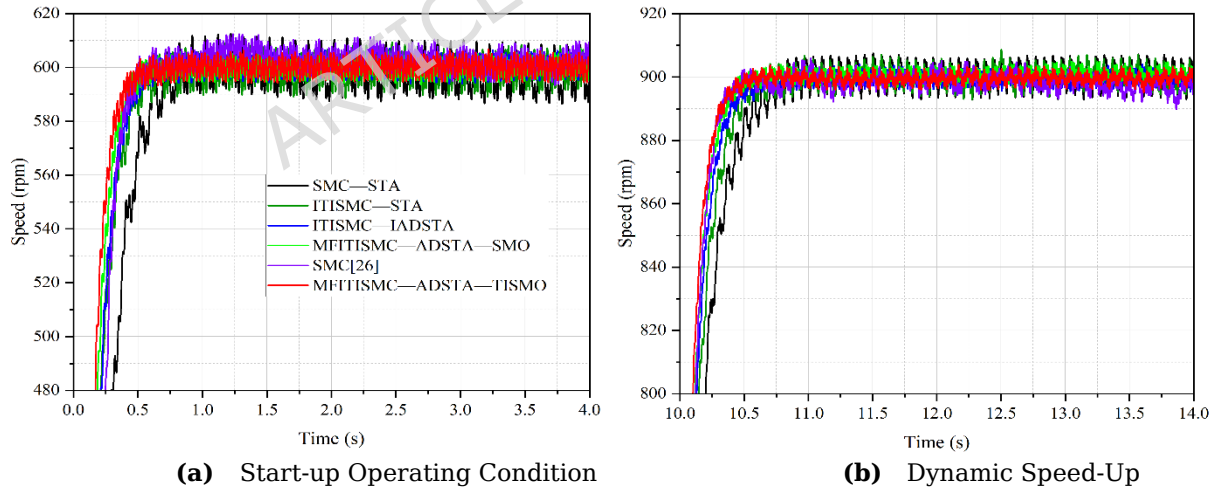
As shown in Fig. 6(g), a sinusoidal disturbance of amplitude $0.18 \tau_m^{-1}$ (where τ_m denotes the mechanical time constant) is applied at 35 s. SMC-STA has a 10 r/min fluctuation amplitude with obvious speed deviation, as its fixed gain cannot adapt to periodic time-varying disturbances. ITISM-C-STA reduces the amplitude to 8 r/min by improving time-varying disturbance tracking. ITISM-C-ADSTA lowers it

to 6 r/min via disturbance-period-matched adaptive gain adjustment. MFITISM-C-ADSTA-SMO cuts it to 5 r/min through SMO preliminary disturbance compensation. MFITISM-C-ADSTA-TISMO achieves the lowest amplitude of 4 r/min, delivering optimal periodic disturbance suppression via ADSTA dynamic matching, ultra-local model robustness and TISMO residual disturbance compensation.

7. Flux Linkage Increased to 1.3 Times Its Initial Value

Fig. 6(h) shows the speed response when flux linkage suddenly rises to $1.3 \times \text{rated}$ at 50 s. SMC-STA has a 39 r/min speed drop and 0.75 s recovery time, with degraded performance from model mismatch. ITISM-C-STA reduces the drop to 37.5 r/min and time to 0.6 s via enhanced robustness. ITISM-C-ADSTA further lowers the drop to 32 r/min and time to 0.55 s through ADSTA adaptive adjustment. MFITISM-C-ADSTA-SMO cuts the drop to 29 r/min and time to 0.35 s via model-free design and SMO compensation. MFITISM-C-ADSTA-TISMO performs best with a 25 r/min drop and 0.3 s recovery time, eliminating model mismatch by treating flux perturbations as unknown disturbances and enabling fast TISMO observation and compensation, showing excellent parameter robustness.

Compared with the SMC method in Ref. [26] under full operating conditions, the proposed MFITISM-C-ADSTA-TISMO shortens start-up and dynamic speed regulation response time by 15.38%, achieving zero overshoot/undershoot for faster and stabler tracking. Steady-state speed fluctuation is reduced by 33.3%, with significantly improved chattering suppression. Under step load disturbance, speed fluctuation drops by 17.95% and recovery time shortens by 25%. Under $0.18\tau_m^{-1}$ sinusoidal disturbance, fluctuation amplitude decreases by 27.27% with smaller phase lag. Under flux linkage perturbation, speed drop is reduced by 16.67% and recovery time shortened by 45.45%, realizing comprehensively improved parameter robustness.



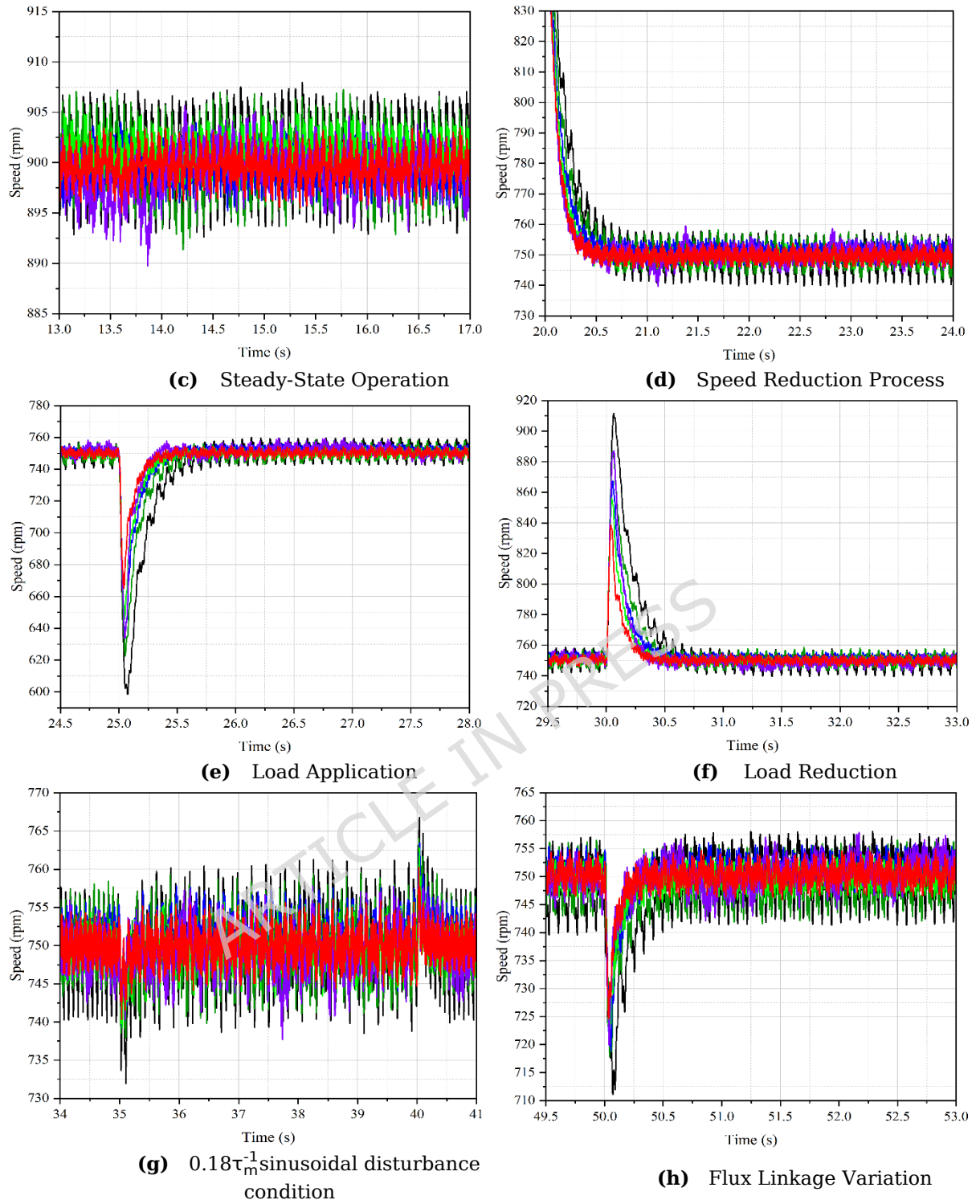


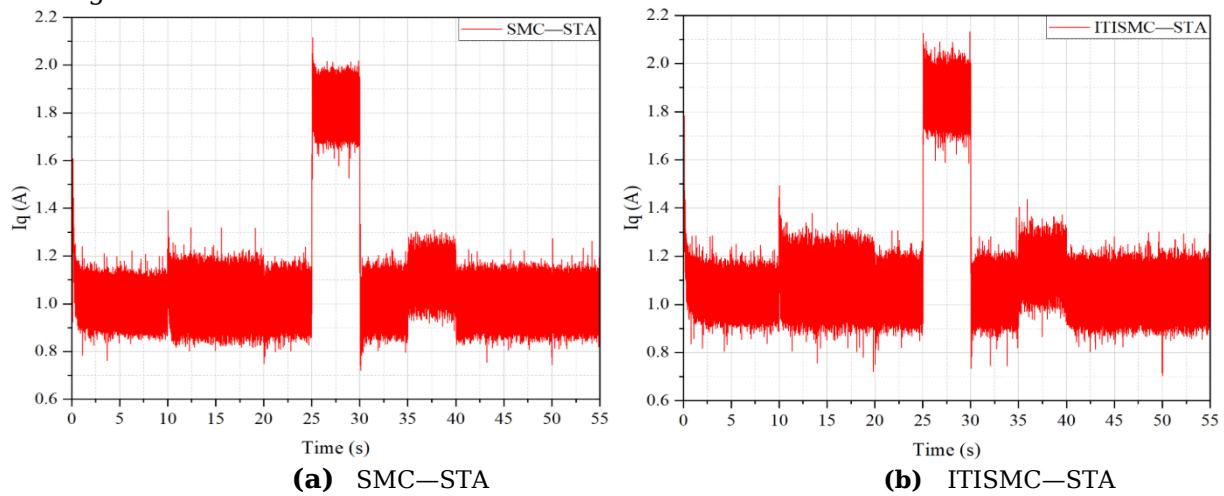
Figure 6. Speed Experiments

Experimental results across all operating conditions demonstrate that the proposed improved TISMO achieves excellent disturbance-handling performance. Under abrupt load changes, sinusoidal disturbances and parameter perturbations, the system exhibits smaller speed fluctuations, faster disturbance recovery and smoother q-axis current. These results confirm that the TISMO effectively estimates and compensates for unknown disturbances in real time, providing strong support for improved system disturbance rejection.

Table 1. Performance Comparison of Different Control Methods

Control Method	No-load Start-up Response Time (s)	Rise Time to 900 rpm (s)	Chattering (Speed Fluctuation, r/min)	Flux change (Acceleration, s)	Load Application (Speed Fluctuation, r/min)	$0.18\tau_m^{-1}$ Sinusoidal Disturbance (Speed Fluctuation, r/min)
SMC—STA	0.8	0.85	7	0.7	150	10
ITISMC—STA	0.7	0.7	6	0.65	128	8
ITISMC—ADSTA	0.65	0.65	5	0.5	119	6
MFITISMC—ADSTA—SMO	0.6	0.6	4	0.45	115	5
SMC[26]	0.65	0.55	4.5	0.55	117	5.5
MFITISMC—ADSTA—TISMO	0.55	0.5	3	0.3	96	4

To further verify the chattering suppression performance of different control methods, the experimentally obtained q-axis current waveforms are presented in Fig. 7.



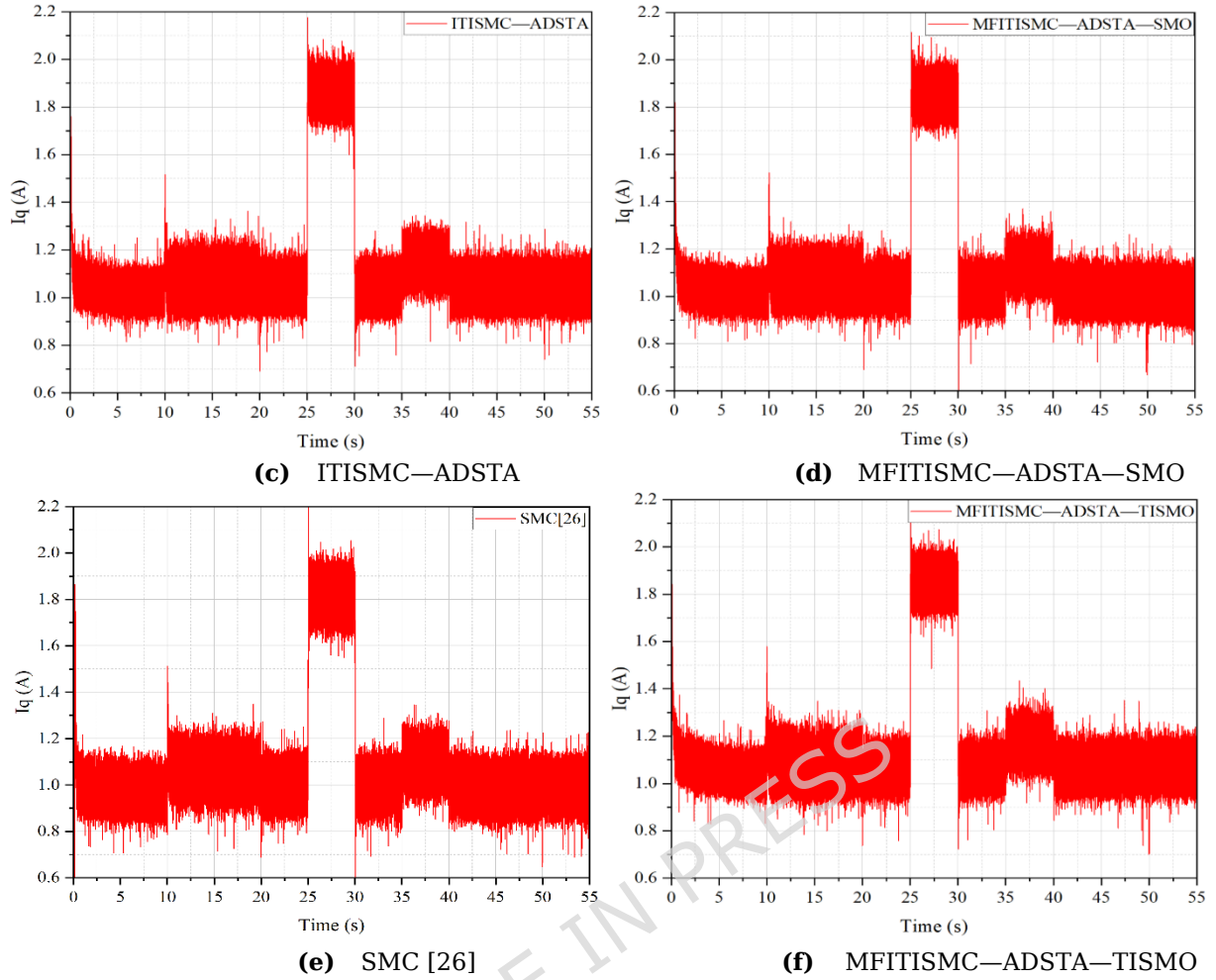


Figure 7 q-axis Current Experiments

The q-axis current waveforms gradually optimize from SMC-STA to MFITISM-ADSTA-SMO. SMC-STA exhibits the severest chattering with dense high-frequency noise; under 25 s abrupt load change and dynamic conditions, its current fluctuates widely at 1.67–1.99 A, due to the sign function's inherent chattering and fixed-gain reaching law's poor adaptability. ITISM-STA reduces chattering and noise via a terminal integral sliding-mode surface, slightly improving current smoothness but still having periodic fluctuations and limited disturbance rejection. ITISM-ADSTA further weakens chattering with an ADSTA (time-varying gain adapting to working conditions), narrowing current fluctuation to ~1.71–2 A, yet it relies on accurate motor models, leading to insufficient disturbance compensation. MFITISM-ADSTA-SMO integrates a model-free design and a conventional SMO, further reducing fluctuations, with current chattering at 1.71–1.98 A, but residual disturbances from limited SMO accuracy cause minor fluctuations.

The proposed MFITISM-ADSTA-TISMO achieves optimal q-axis current performance: minimal chattering and noise, steady-state fluctuation of 1.73–1.97 A under abrupt load change and dynamic conditions, and the fastest recovery. This is attributed to its novel ultra-local model (unifying parameter perturbations and external disturbances), improved TISMO for accurate real-time disturbance estimation and compensation, and synergistic ADSTA adaptive reaching law and model-free feature that fundamentally suppresses chattering, verifying its

superiority in current-loop control. Compared with the SMC method in Ref. [26], the proposed MFITISM-C-ADSTA-TISMO strategy achieves significant suppression of q-axis current chattering. Under a load step at 25 s, the current fluctuation is reduced by 0.1 A relative to SMC [26], while the steady-state current is smoother with a reduction of 0.12 A, demonstrating superior control performance.

6 Conclusion

Aiming at the problems of conventional sliding-mode control in high-precision control scenarios of permanent magnet synchronous motors (PMSMs), such as high-frequency chattering, slow dynamic response, insufficient anti-disturbance capability and poor parameter robustness, this paper proposes a model-free sliding-mode control strategy (MFITISM-C-ADSTA-TISMO) based on improved terminal integral sliding-mode and adaptive super-twisting reaching law. A novel ultra-local model and an improved terminal sliding-mode observer (TISMO) are constructed to achieve accurate estimation and compensation of unknown system disturbances. The speed-loop and q-axis current control performances of the conventional sliding-mode SMC-STA, ITISM-C-STA, ITISM-C-ADSTA, MFITISM-C-ADSTA-SMO and the proposed strategy are systematically compared under the operating conditions of no-load startup, dynamic speed increase, steady-state operation, dynamic speed reduction, load abrupt change, $0.18\tau_m^{-1}$ sinusoidal disturbance and flux linkage parameter variation. Experimental results demonstrate that the proposed MFITISM-C-ADSTA-TISMO exhibits comprehensive advantages in dynamic response speed, steady-state control accuracy, high-frequency chattering suppression, multi-scenario anti-disturbance performance and parameter robustness. It provides a novel technical approach for high-performance control of PMSMs.

The proposed MFITISM-C-ADSTA-TISMO controller achieves collaborative innovation through multiple modules, and its core advantages can be summarized as follows:

- 1□ Chattering Suppression and High Steady-State Accuracy: By abandoning the discontinuous switching characteristic of the conventional sign function, a continuous smooth nonlinear function is adopted to construct the terminal integral sliding-mode surface. Combined with the adaptive super-twisting reaching law (ADSTA), the generation source of high-frequency sliding-mode chattering is eliminated at the mechanism level. The novel ultra-local model further weakens the influence of model errors and small disturbances on the system, significantly improving the steady-state control accuracy of the speed loop and the smoothness of the q-axis current, and thoroughly solving the inherent chattering problem of conventional sliding-mode control.
- 2□ Fast Dynamic Response and Precise Tracking Ability: The ADSTA adaptive reaching law can dynamically adjust the reaching gain according to the system operating conditions. Combined with the terminal integral sliding-mode surface with finite-time convergence, the transient response speed and regulation stability of the speed loop are greatly enhanced. It can realize real-time and precise tracking of the speed reference value under dynamic speed-up, speed-reduction and other working conditions without obvious overshoot and undershoot, solving the problems of slow dynamic response and insufficient tracking accuracy in conventional sliding-mode control.

- 3□ Strong Anti-Disturbance Performance and Parameter Robustness: The improved TISMO realizes accurate real-time estimation and compensation of various unknown disturbances such as load abrupt change, sinusoidal disturbance and flux linkage parameter perturbation. Combined with the model-free design, it completely gets rid of the dependence on accurate motor parameters. Under complex disturbance conditions, it can quickly suppress speed and current fluctuations and ensure the system returns to steady state rapidly. It effectively solves the pain points of conventional sliding-mode control, such as dependence on accurate models and weak anti-disturbance ability, providing a solid guarantee for the stable and reliable operation of permanent magnet synchronous motors under complex working conditions.

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Author contributions

Y.W.: Conceptualization, Methodology, Validation, Writing – Original Draft. W.L.: Supervision, Funding acquisition, Project administration, Writing – Review & Editing. All authors reviewed and approved the final manuscript.

Data availability statement

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare no competing interests.

7 References

1. Vu Q H, Vu T L, Duong M T, et al. Comparison of Various Permanent Magnet Synchronous Motor Concepts Used for Electric Ship Application[C]//2025 28th International Conference on Electric Machines and Systems (ICEMS). IEEE, 2025: 725-730.
2. Zwierzewicz Z, Tarnapowicz D, Nerc A. Adaptive PMSM Control of Ship Electric Electric with Energy-Saving Features[J]. Energies, 2024, 17(9): 2178.
3. Zhang, L.; et al. PMSM non-singular fast terminal sliding mode control with disturbance compensation. Inf. Sci. 2023, *642*, 119040.
4. Z. Wang, S. Lu, S. Zheng, M. Zhang, and B. Song, "Anti-saturation speed regulation using dual-parameter sliding mode observer for PMSM servo systems", IEEE Trans. Ind. Electron., pp. 1–12, 2026,
5. Wang Z, Lu S, Zheng S, et al. Lumped Disturbance and Moment of Inertia Identification for Servo Speed Regulation Based on Antisaturation Finite-Time Extended State Observer[J]. IEEE Transactions on Power Electronics, 2026.
6. Ma X, Zhang L, Xu Y, et al. Model-free control of permanent magnet synchronous motors with improved terminal sliding mode based on a modified exponential reaching law[J]. Control Engineering Practice, 2026, 166: 106630.
7. Wu W, Huang S, Qin J, et al. Design of a fractional-order sliding mode controller for lane-keeping in autonomous driving[J]. Scientific Reports, 2025, 15(1): 33602.

8. Hou L, Lan X. Fixed-time consensus multi-agent-systems-based speed cooperative control for multiple permanent magnet synchronous motors with complementary sliding mode control[J]. *Electronics*, 2024, 13(22): 4407.
9. Wang X, Wang B, Wu Y, et al. Adaptive control and disturbance compensation for gear transmission servo systems with large-range inertia variation[J]. *Transactions of the Institute of Measurement and Control*, 2022, 44(3): 700-715.
10. Calgan H. Novel tilt integral sliding mode controller and observer design for sensorless speed control of a permanent magnet synchronous motor[J]. *COMPEL-The international journal for computation and mathematics in electric and electronic engineering*, 2022, 41(1): 455-470.
11. Pei H, Fan P, Ma X, et al. A fast adaptive sliding mode observer for permanent magnet synchronous motors based on new segmented reaching law[J]. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*, 2025: 09544070251387379.
12. Karboua D, Mebkhoua T, Benkaihou S, et al. Model reference adaptive system and pseudo-sliding mode control with exponential reaching law for sensorless-speed control of PMSM[J]. *Plos one*, 2025, 20(5): e0321985.
13. Zhang L, Tao R, Bai J, et al. An improved sliding mode model reference adaptive system observer for PMSM applications[J]. *Expert Systems with Applications*, 2024, 250: 123907.
14. Hu M, Ahn H, Chung Y, et al. Speed regulation for PMSM with super-twisting sliding-mode controller via disturbance observer[J]. *Mathematics*, 2023, 11(7): 1618.
15. Shi L, Lv M, Li P. Sensorless position control in high-speed domain of PMSM based on improved adaptive sliding mode observer[J]. *Processes*, 2024, 12(11): 2581.
16. Wang X, Liu Z, Zhou P, et al. A Saturation Adaptive Nonlinear Integral Sliding Mode Controller for Ship Permanent Magnet Electric Motors[J]. *Journal of Science and Engineering*, 2025, 13(5): 976.
17. Rezaei A, Tabatabaei M. Ship roll stabilization using an adaptive fractional-order sliding mode controller[J]. *Ocean Engineering*, 2023, 287: 115883.
18. Bai H, Yu B. Position estimation of fault-tolerant permanent magnet motor in electric power electric ship system[J]. *IEEE Transactions on Electric and Electronic Engineering*, 2022, 17(6): 890-898.
19. Zwierzewicz Z, Tarnapowicz D, German-Galkin S, et al. Optimal control of the diesel-electric electric in a ship with PMSM[J]. *Energies*, 2022, 15(24): 9390.
20. Zhu Z, Chang L, Pan S. Simulation Rerch on Ship-borne PMSM Speed Regulation Control System Based on Fuzzy PID Control[C]//*Journal of Physics: Conference Series*. IOP Publishing, 2022, 2417(1): 012036.
21. Karakatsanis T S. Optimization and Performance Evaluation of PM Motor and Induction Motor for Electric Systems[J]. *Applied System Innovation*, 2025, 8(3): 58.
22. Zhang L, Tao R, Li X. Model-Free Adaptive Second Order Sliding Mode Control for PMSM in Electric Traction Systems Based on Terminal Attractor[J]. *IEEE Transactions on Power Electronics*, 2026.
23. Hua C, Shi D, Chen X, et al. Ultra-Local Model-Based Adaptive Enhanced Model-Free Control for PMSM Speed Regulation[J]. *Machines*, 2025, 13(7): 541.
24. Tu L X, Zhao K, Fang B, et al. A Novel Model-Free Adaptive Sliding Mode Control of PMSM Based on Dynamic Linearization[J]. *International Journal of Circuit Theory and Applications*, 2025.

25. Zhong Y, Lin H, Chen Z, et al. Online-parameter-estimation-based control strategy combining MTPA and flux-weakening for variable flux memory machines[J]. IEEE Transactions on Power Electronics, 2021, 37(4): 4080-4090.
26. Zhao K, Liu W, Zhou R, et al. Model-free fast integral terminal sliding-mode control method based on improved fast terminal sliding-mode observer for PMSM with unknown disturbances[J]. ISA transactions, 2023, 143: 572-581.
27. Ullah S, Khan Q, Mehmood A, et al. Neuro-adaptive fast integral terminal sliding mode control design with variable gain robust exact differentiator for under-actuated quadcopter UAV[J]. ISA transactions, 2022, 120: 293-304.
28. Li T, Sun X, Su Z, et al. Model-free predictive control for harmonic suppression of PMSMs based on adaptive resonant controller[J]. IEEE Transactions on Energy Conversion, 2025.
29. Li T, Sun X, Yao M, et al. Improved finite control set model predictive current control for permanent magnet synchronous motor with sliding mode observer[J]. IEEE Transactions on Transportation Electrification, 2023, 10(1): 699-710.
30. Li T, Sun X, Yang Z, et al. Simplified two-step model predictive control with fast voltage vector rch[J]. IEEE Transactions on Industrial Electronics, 2024, 72(4): 3303-3312.
31. Su Z, Sun X, Lei G, et al. Model-free predictive current control for dual three-phase PMSM drives with an optimal modulation pattern[J]. IEEE Transactions on Industrial Electronics, 2023, 71(9): 10140-10149.
32. Su Z, Sun X, Lei G, et al. Improved model-free predictive current control for SPMSM drives with adaptive prediction horizon strategy[J]. IEEE Transactions on Industrial Electronics, 2024, 72(6): 5705-5715.
33. Su Z, Sun X, Ortombina L, et al. Model-Free Predictive Current Control With Virtual Harmonic Back-EMF Identification for Dual Three-Phase PMSM Drives[J]. IEEE Transactions on Industrial Electronics, 2026.